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RESEARCH ARTICLE



Global climate policy with differentiated carbon price floors

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ABSTRACT

We use a state-of-the-art global general equilibrium model to quantify the effects of an International Carbon Price Floor (ICPF) proposal for the coordination of global climate policy. With the ICPF proposal, countries would implement a carbon price floor that is differentiated according to their level of development. The price floor could be implemented by an actual carbon price or other policies with equivalent stringency. Implementing such a joint effort would be a big step towards aligning global emissions with the Paris Agreement. The study highlights three main findings. First, the ICPF is progressive in terms of emission reductions and demands considerably more effort from high-income countries (HICs). Unlike proposals based on equity principles, however, it requires neither extremely high emission reductions from HICs nor massive financial transfers. Second, the differentiated carbon prices cost only an additional 0.3 percent of global GDP compared to a global uniform carbon price. Third, the ICPF can be as effective as border carbon adjustment (BCA) mechanisms at preserving competitiveness of HICs despite the differentiation in carbon price floors. This means, there would be no need for BCA with an ICPF. Compared to a BCA, HICs can avoid the administrative burden and low-income countries (LICs) will not face the cost of BCA. As a first step to global climate policy, a sectoral ICPF for the EITE sector could be introduced. It would address competitiveness concerns almost as well as the full ICPF.

Key policy highlights:

- An International Carbon Price Floor (ICPF), with carbon price floors differentiated by development level, would incur only small economic costs compared to baseline economic growth.
- The ICPF would allow for international burden sharing, with HICs incurring higher cost than middle – and low-income countries.
- The ICPF reduces distortions to competitiveness as much as BCA, so that no BCA would be necessary.
- A ‘sectoral ICPF’ in energy-intensive and trade-exposed sectors would be effective against leakage and loss in competitiveness and would thus be an attractive alternative to border carbon adjustment.

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1. Introduction

Concerns about compatibility of decarbonization with continued robust growth, fairness of international burden sharing, and risks of competitiveness losses are obstacles to enhanced international climate action. For low-income countries (LICs) and middle-income countries (MICs), the resistance to more climate action is rooted in the concern that decarbonization would compromise their development prospects by making energy more expensive, given their fast-growing energy needs. LICs and MICs also highlight that high-income countries (HICs) should bear a larger burden of the global mitigation effort given their responsibility

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for the large majority of historical greenhouse gas (GHG) emissions. In contrast, HICs are concerned about competitiveness losses for their energy-intensive and trade-exposed (EITE) industries and carbon leakage if they adopt more stringent carbon-pricing policies than their partners. However, international cooperation to motivate action by all country groups is key to controlling climate change. HICs alone cannot reduce global emissions sufficiently, given their declining projected share in future emissions. In addition, MICs and LICs have a strong interest in a successful global effort to limit climate change, as they are highly vulnerable to climate change.

As a way forward, the IMF recently proposed introducing an international carbon price floor (ICPF) arrangement among a small number of large emitters covering the bulk of global emissions to scale up action on mitigation. The proposal is organized around carbon pricing but also allows for alternative policy approaches with similar effects on emissions (Parry et al., 2021). Guided by the principle of common but differentiated responsibilities of the Paris Agreement, the proposal uses carbon price floors that are differentiated by the level of development of countries and includes the possibility of complementing the price floors with financial and technology assistance to LICs. The idea of using global carbon prices is not new, but what differentiates some of these recent proposals is the idea of using carbon price floors – which allow more ambitious countries to do more – and to differentiate these by income levels. The agreement on joint minimum carbon prices could dispense with the need for border carbon adjustment (BCA) among the participating countries, although the agreement could still be supported by a BCA on nonparticipants.

This paper quantifies the effects of the ICPF proposal, discussing its emission, economic, burden-sharing, and competitiveness impacts, and assesses it against two major alternatives – a uniform carbon price and unilateral action by HICs with border carbon adjustment. The ICPF scenario is an international carbon price floor, in which high-, middle- and low-income countries introduce carbon price floors reaching US\$75, US\$50, and US\$25 by 2030, respectively. Countries implement the maximum of their carbon price floor and the carbon price implicit in their Nationally Determined Contribution. The three price floors have the advantages of reducing emissions considerably, being progressive in the sense of asking most of the most advanced economies and being simple and transparent.

The reference scenario calculates the uniform global carbon price, assumed to be implemented by all countries, which delivers cumulative emissions reductions similar to those of the international carbon price floor arrangement. In this scenario, the carbon price in 2030 is about US\$56. Third, a set of partial action scenarios assume that MICs and LICs do not want to implement an economy-wide international carbon price floor and, as a consequence, are subject to a border carbon adjustment. Finally, the paper explores an international carbon price floor arrangement limited to energy-intensive and trade-exposed sectors, which would avoid the imposition of border carbon adjustments. This study uses a newly developed dynamic computable general equilibrium model named ‘IMF-ENV’. IMF-ENV is well suited for analyzing large and simultaneous policy changes across the global economy. It integrates trade impacts and induced changes in global energy prices. Finally, it provides a broad assessment of economic impacts on GDP, employment, investment, competitiveness, and burden sharing.

Many approaches to climate policy coordination are based on conceptual equity principles. Past research discusses the implications of different criteria for fairness of international burden sharing of climate mitigation (Clarke et al., 2014; Rose et al., 1998; van den Berg et al., 2020). Implementing equity principles like responsibility, capability and equal cumulative per capita emissions through country-specific emission reductions would involve very stringent emission reductions for HICs (Höhne et al., 2014). Implementing global decarbonization instead through a global carbon price would require sizeable financial transfers from HICs to comply with the equity principles (Leimbach & Giannousakis, 2019).

This paper follows a more ‘pragmatic’ approach in the sense of not following one of the equity principles and does not require either very large transfers or extremely stringent emission reductions by HICs. It shows that moderately differentiating carbon prices can go a substantial way toward making the burden sharing of global mitigation efforts more progressive at moderate global efficiency costs. Achieving full progressivity of economic costs would, however, require additional climate finance in accordance with Article 9 of the Paris Agreement.

Other studies have focused on practical aspects of climate policy coordination. For example, Nordhaus (2015) finds that climate policy coalitions are not stable unless they use sanctions, like small trade penalties.

The stability is based on a comparison of the economic benefits of being inside or outside a climate coalition. This paper is based on the premise that countries do see climate policy as being in their self-interest but want global cooperation to be equitable. This is achieved by balancing the interests of different income groups. Thube et al. (2021) emphasize the benefits of extending climate policy as broadly as possible, to countries, greenhouse gases and sectors. The ICPF seeks to take advantage of this efficiency gain by applying climate policy to all emissions globally.

The most similar paper to this might be Bauer et al. (2020), which finds that a combination of moderately differentiated carbon prices and a moderate level of financial transfers achieves equity without unrealistic demands to HICs. This paper also includes trade differentiated by various end-use sectors that incorporate CO₂ emissions due to energy inputs. In addition, this paper evaluates differences in distributional outcomes of differentiated carbon pricing, whereas Bauer et al. (2020) impose the distributional outcome as a constraint and vary the carbon price differentiation and the financial transfers to meet it for a given climate target.

This paper contributes to the literature that examines fragmented climate mitigation approaches and the role of border carbon adjustment in reducing carbon leakage and competitiveness losses for acting countries. The literature finds that a border carbon adjustment helps reduce competitiveness losses of energy-intensive and trade-exposed sectors (see for example Böhringer et al., 2022; Burniaux et al., 2013). Border carbon adjustments are, however, not a perfect instrument to restore output levels of energy-intensive and trade-exposed industries (Thube et al., 2021). Compared with this existing literature, this paper analyzes the role of an ICPF to reduce competitiveness losses and the possibility of a sectoral climate agreement.

The paper is structured as follows. Section 2 presents the model and introduces the scenarios. Section 3 presents the effects of an ICPF on emissions, economic development, international burden sharing, and competitiveness, and compares it to other scenarios. Section 4 concludes.

2. Methods

The analysis in this paper uses the new IMF-ENV model, a dynamic computable general equilibrium (CGE) model. Further details of the model and a description of the baseline scenario are provided in Appendix 1.

2.1 The IMF-ENV model

IMF-ENV is a CGE model newly developed by the IMF Research Department. It is built on the ENVISAGE model (van der Mensbrugghe, 2019) and the OECD ENV-Linkages Model (Château et al., 2021). Dynamic CGE models are well adapted for the analysis of policies implying structural change (i.e. changes in the sectoral composition of economies) like those resulting from ambitious decarbonization goals. These models are based on a neoclassical framework, dealing only with real values and with almost-perfect markets for goods and production factors. They focus on the long-term reallocation of resources across the different sectors and allow the simulation of impacts of climate mitigation policies on emissions, macroeconomic variables, sectoral economic activity, and international trade patterns. They feature vintage capital stocks and short-term adjustment costs for the capital stock but tend to underestimate some of the short-term costs of the transition, for example, those resulting from rigidities in labour reallocation.

For the sake of exposition, the 25 regions and countries considered in the model are grouped into four categories in most of the paper's figures, namely, HICs, MICs, and LICs (following World Bank classification) and oil exporters, see Table 1.

2.2 Policy scenario overview

The baseline scenario projects economic development and emissions under the assumption that climate mitigation is limited to the current actual climate and energy policies. Based on recent trends, energy and emission intensities are projected to decline due to a combination of 'electrification' of the energy system, the progressive increase in importance of renewable energy in the electricity mix, and improvements in sectoral energy efficiency. These trends, however, are insufficient to offset the effects of GDP growth on emissions, implying

Table 1. Country groups in IMF-ENV model.

High-income countries	Middle-income countries	Low-income countries
Australia	Argentina	India
Canada	Brazil	Other Africa (OAF)
France	China	Other East Asia and New Zealand (ODA)
Germany	Indonesia	Other Europe and Asia (OEURASIA)
Italy	Mexico	
Japan	South Africa	
Republic of Korea	Other Latin America (OLA)	Oil-exporting countries^a
Rest of European Union, Iceland, Norway, and Switzerland (RESTEU)	Turkey	Russian Federation
United Kingdom		Saudi Arabia
United States		Other oil-exporting countries (RESTOPEC)

^aOil-exporting countries are separated for convenience. In terms of the economic development classification, Saudi Arabia belongs to the high-income countries group, the Russian Federation to the middle-income countries group, and 'Other oil-exporting countries' to the low-income countries group.

that GHG and CO₂ emissions grow by 23.7 and 23 percent, respectively, over 2019–2030 (see Appendix 1 for more details on assumptions).

The paper builds on this baseline to analyze the ICPF proposal and compares it with a global uniform carbon price and partial-action scenarios with BCAs. Table 2 provides an overview of the scenarios. A full description is contained in the Supplementary Material. The scenarios are very different in the emission reductions they achieve. Only the ICPF and the global tax scenario reduce emissions strongly. Importantly, the current NDCs are far from sufficient to achieve the 2°C target (Black et al., 2021). In all policy scenarios, the revenue from carbon pricing is assumed to be used to reduce distortionary labour income taxes, as is standard in the double-dividend literature.

3. Results

3.1 Implemented carbon prices

In the ICPF scenario, countries implement the maximum of the price floor corresponding to their development level and the price necessary to reach their NDC. Figure 1 provides an overview of the carbon prices for the model regions over time. The carbon prices in some MICs (Argentina, Brazil, and Mexico) and all HICs except Australia are higher than the carbon price floor for their country groups. Reaching NDCs requires carbon prices close to US\$225 in HICs on average, well above the carbon price floor of US\$75, reflecting the high

Table 2. Scenario overview.

Name	Description
Baseline	Countries do not implement additional climate policies.
ICPF	Global ICPF in which countries gradually implement a carbon price up to 2030 to reach the maximum of their carbon price floor and the carbon price implicit in their NDC. Carbon price floors are US\$25/tCO ₂ eq for LICs, US\$50 for MICs, and US\$75 for HICs.
Global tax	Global carbon tax, uniform across all sources and countries, to meet exactly the same global GHG emissions reduction target as in ICPF.
HICs only	Only the HICs country group and Saudi Arabia (which is also a HIC) are acting, implementing their carbon price in the ICPF scenario; MICs and LICs do not implement additional climate policies (not even to fulfil NDCs).
HICs with BCA	Only HICs and Saudi Arabia are acting, implementing their carbon price in the ICPF scenario as well as a BCA. MICs and LICs do not implement additional climate policies. The BCA can be introduced in a standard or maximum way and on all goods or only EITE goods (see text).
EITE	Only HICs and Saudi Arabia are acting, implementing their carbon price in the ICPF scenario. MICs and LICs adopt their carbon price in the ICPF scenario only for EITE industries.

Source: Authors.

Note: BCA = border carbon adjustment; EITE = energy intensive and trade exposed; GHG = greenhouse gas; HIC = high-income country; ICPF = international carbon price floor; LIC = low-income country; MIC = middle-income country; NDC = Nationally Determined Contribution; tCO₂eq = ton(s) of carbon dioxide equivalent.

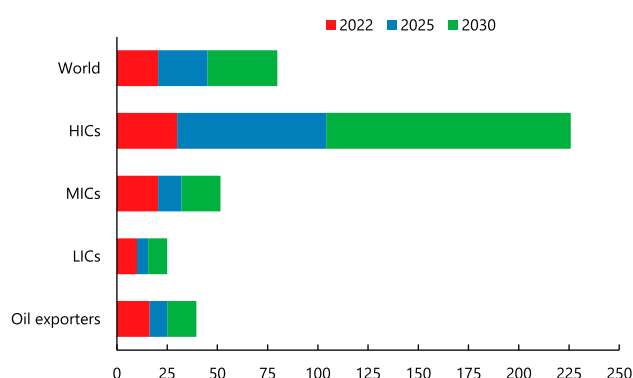


Figure 1. Carbon prices under the ICPF scenario, by aggregate country groups. Sources: Price floors from Parry et al. (2021) and IMF-ENV model for endogenous carbon price to reach Nationally Determined Contribution (NDC) target; NDC targets from Climate Watch (2021). Note: HIC = high-income country; LIC = low-income country; MIC = middle-income country; tCO₂eq = ton(s) of carbon dioxide equivalent.

ambition of their commitments. The left panel of Figure 3 (below) shows for which countries the ICPF is binding and implies emission reductions larger than their NDC.

Figure 9 in Appendix 2 show the carbon prices in the ICPF scenario by country. It also marks the carbon prices necessary to implement the NDCs. By definition of the ICPF scenario, the ICPF carbon prices for countries with ambitious NDCs coincide with the carbon prices needed to reach the NDCs. The highest carbon prices would be required by France (US\$ 380), Canada (US\$ 340) and the UK (US\$ 340). IMF-ENV is closely related to the ENVISAGE model, so that its carbon prices needed to implement the NDCs are a bit above median among the models compared in (Böhringer et al., 2021, Fig. 2).

The study estimates that achieving NDCs with carbon prices alone could require very high carbon prices by 2030 in some countries. Carbon prices measure marginal abatement costs, that is, the cost of removing the last unit of carbon, which is typically higher for countries with very ambitious NDCs (e.g. the United States) and for countries with very low current levels of emissions (e.g. France). This is because to reach their ambitious targets, they must decarbonize hard-to-abate sectors such as industry and transportation.¹ In a review of model estimates of carbon prices consistent with the 2°C target, the High-Level Commission on Carbon Prices (2017) found estimated carbon prices in 2030 ranging from US\$50 to US\$100. The corresponding uniform carbon price in this study's global tax scenario (presented later in the paper) is US\$55.80.

3.2 The effect of the iCPF on projected GHG emissions

In the ICPF scenario, future emissions fall strongly compared with the baseline, making considerable progress towards the 2°C target. Results in the top left panel of Figure 2 show that in 2030, global GHG emissions fall by 26.6 percent relative to their baseline levels (29 percent for CO₂ emissions, not shown). However, because emissions are projected to increase strongly in the baseline, this translates into more modest reductions of 9 percent from 2019 levels (13 percent for CO₂ emissions). These emission reductions put global emissions on a declining trend in a context of robust global growth. The ICPF and the global tax scenarios have significantly larger emission reductions than the partial action scenarios, see Figure 10 in Appendix 2.

Emission reductions in HICs are the largest, followed by those in MICs and then those in LICs and oil-exporting countries. HICs reduce GHG emissions by 45 percent (and CO₂ emissions by 43 percent) in 2030 compared with the baseline (top left panel of Figure 2). This reduction is higher than that in other countries, because as Figure 1 shows, the NDCs of most HICs imply emission reductions beyond the US\$75/tCO₂eq required by the carbon price floor.

MICs reduce GHG emissions by about 25 percent relative to their baseline (and CO₂ emissions by 28 percent). This is more than these countries would do to achieve their NDCs (Figure 3). There is considerable within-group variation, as a given carbon price reduces emissions more among countries with a higher carbon intensity.

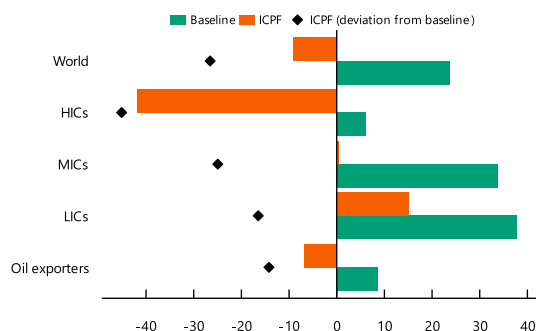
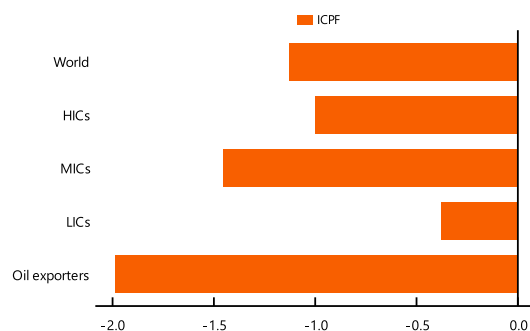
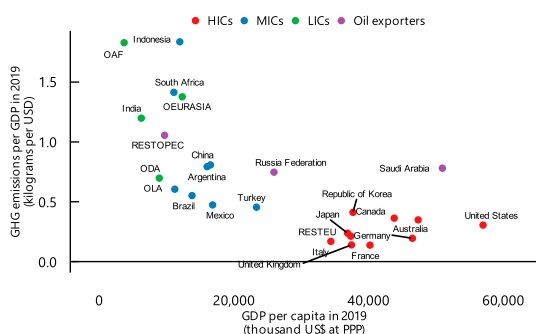
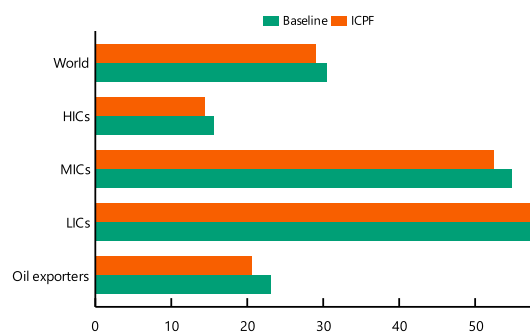
GHG emissions in 2030*(percent change from 2019)***Real GDP in 2030***(percent deviation from baseline)***GHG emissions per GDP versus GDP per capita in 2019****Real GDP in 2030***(percent change from 2019)*

Figure 2. ICPF scenario: Changes in GHG emissions and GDP in 2030, by aggregate country groups. Source: IMF-ENV model. Note: GHG = greenhouse gas; HIC = high-income country; ICPF = international carbon price floor; LIC = low-income country; MIC = middle-income country; OAF = Other Africa; ODA = Other East Asia and New Zealand; OEURASIA = Other Europe and Asia; OLA = Other Latin America; PPP = purchasing power parity; RESTEU = Rest of European Union and Iceland; RESTOPEC = Other oil-exporting countries. GHG emissions contains CO₂ LULUCF.

China, Indonesia, and South Africa have a high carbon intensity and drive the relatively high emission reductions of MICs (left panel of Figure 3).

LICs and oil producers reduce their GHG emissions less than the other groups, by 16.5 and 14.2 percent, respectively, relative to their baselines, reflecting their low carbon price floors. Many LICs, however, have NDCs, which would require even less emission reductions, as is the case for MICs. Despite reductions of emissions relative to the baseline, all LICs experience rising emissions relative to 2019 under the ICPF. Among LICs, India is playing a central role, since it accounts for 35 percent of GHG emissions of this group in 2030. Specifically, GHG emissions in India decline by 20 percent compared with the baseline in 2030 and increase by 19.4 percent compared with 2019 emissions. The share of global GHG emissions for India would increase from 7.2 percent to 9.5 percent between 2019 and 2030 (under the ICPF).

3.3 The effect of the iCPF on economic development

The aggregate economic costs under the ICPF are small, considering the large emission reductions from baseline. The model projects a decrease in global real GDP of about 1.1 percent in 2030 compared with its baseline level, against global GHG emission reductions of 29 percent from baseline in the same year (top left panel of Figure 2). GDP costs for HICs are generally moderate, even though their NDCs imply higher proportional

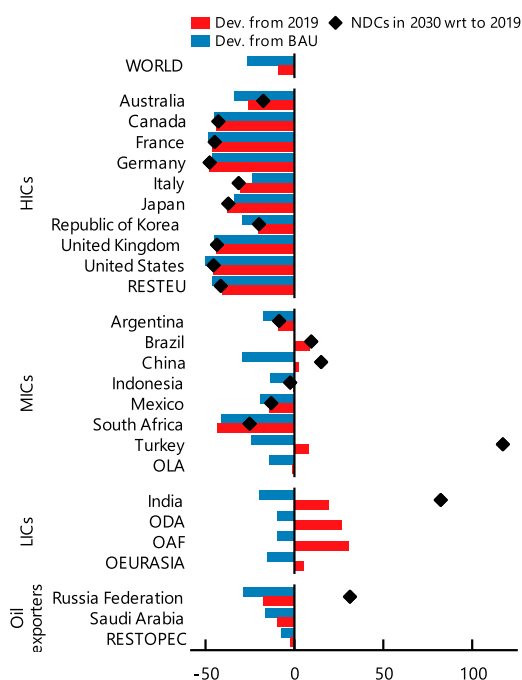
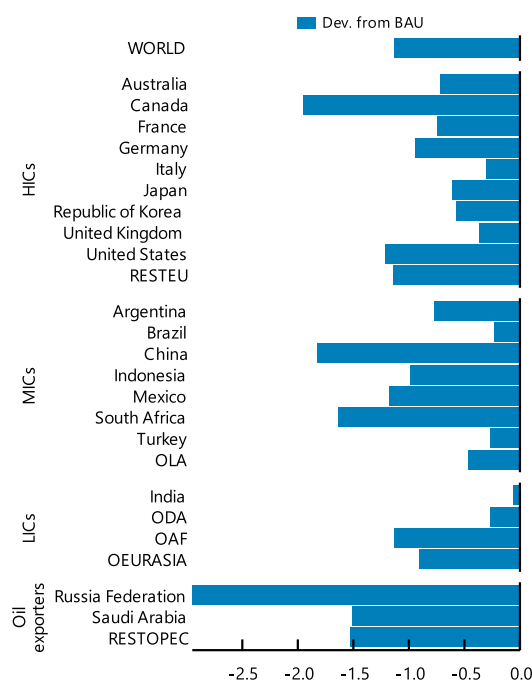
GHG emissions and NDCs¹ in 2030*(percent change)***Real GDP in 2030***(percent change from baseline)*

Figure 3. ICPF scenario: Changes in GHG emissions and real GDP in 2030, by country. Sources: IMF-ENV model; Nationally Determined Contribution (NDC) targets calculated using November 2021 data from Climate Watch (2021). Note: GHG = greenhouse gas; BAU = business as usual; Dev. = deviation; HIC = high-income country; LIC = low-income country; MIC = middle-income country; OAF = Other Africa; ODA = Other East Asia and New Zealand; OEURASIA = Other Europe and Asia; OLA = Other Latin America; RESTEU = Other oil-exporting countries; wrt = with respect to. ¹The emission reduction targets for 2030 proposed in NDCs are generally expressed in terms of total greenhouse gases in megatons of CO₂ equivalent for all sources of emissions (including land use, land use change, and forestry, or LULUCF).

emission reductions and that abatement opportunities are more expensive in these countries. This reflects that HICs are on average less emission intensive than MICs and LICs, as shown by GHG intensities reported in the bottom left panel of Figure 2. Hence, for a given carbon price increase, the corresponding increase in production costs is mechanically lower in HICs than in other countries. These lower relative economic costs of climate mitigation for HICs, compared with those for developing countries, which are characterized by higher growth of GDP and higher reliance on fossil fuels, are a typical result of CGE models (OECD, 2009) and integrated assessment models (Clarke et al., 2014).

Detailed results presented in the right panel of Figure 3 show that countries with large fossil-fuel sectors, like Canada, member countries of OPEC, the Russian Federation, and South Africa, record GDP losses of more than 1.5 percent, which is higher than the global average. Losses in these countries are relatively high even with modest climate ambition, because of the strong decline in their export incomes as well as a high emission intensity of these economies. However, the losses are still moderate, at 3 percent of GDP, compared with the baseline for the Russian Federation, the most-affected region. The Russian Federation is more affected than other oil-exporting countries for multiple reasons. First it faces a high minimum carbon price floor. Second, the share of the Russian Federation's manufacturing and energy-intensive industries is higher than that for other oil exporters. Third, the fall in international oil prices under the ICPF scenario makes Russian oil extraction less profitable relative to that in the other oil-exporting countries.

Among other countries, GDP costs range from 0 (India) to 1.8 percent (China) and increase with the carbon intensity of the country and the level of the carbon price. MICs have on average larger GDP losses, but this

masks different situations across countries of this group. China, which currently has a carbon-intensive economy, has a relatively high cost of 1.8 percent of GDP. Mexico and Indonesia have a historically important reliance on fossil fuels and record GDP losses of about 1.2 and 1 percent, respectively. But other countries in this group show GDP losses well below 1 percent. LICs as a group have smaller losses than MICs and HICs, reflecting the fact that their emission reductions are more moderate under the ICPF.

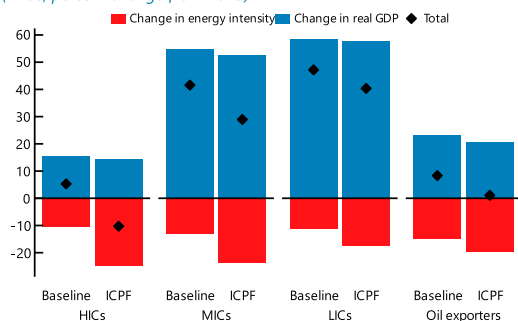
When considering the effect of climate policy on GDP, as in the right panel of Figure 3, it is important to consider that the model does not capture all GDP impacts of climate policy. Climate policy generates substantial co-benefits on output and welfare and also helps reducing future damages from climate change. Figure 12 in Appendix 2 shows the effect of climate policy on air pollutants. The model does not reflect the economic benefits of these reductions, however. The co-benefits from climate policy include the positive effect of climate policy on labour productivity and health through a reduction in local air pollution (Graff Zivin & Neidell, 2012) and the benefits of shifting the tax burden from labour income taxation to carbon pricing in terms of reduced informality of the economy (Bento et al., 2018). In some countries, these effects are projected to offset fully the GDP losses of carbon pricing, for example, because of higher labour productivity under lower air pollution levels (Li et al., 2018; Pandey et al., 2021) and more efficient tax systems (Liu, 2013).

The ICPF is compatible with continued robust growth and development. The bottom right panel of Figure 2 puts GDP costs into perspective, by comparing cumulative growth between 2019 and 2030 in the ICPF and the baseline scenarios. GDP is projected to grow substantially in the baseline: for example, GDP in LICs is expected to grow by 58.3 percent, thus converging toward that in higher-income countries. **The ICPF changes this picture only marginally: cumulative growth is reduced from 30.4 percent to 29.2 percent at the global level and from 58.3 percent to 57.3 percent for LICs.** Effective climate policy is thus not at odds with economic development. Energy consumption keeps growing strongly under the ICPF, especially in MICs and LICs, supporting fast GDP growth (Figure 4). The carbon price provides incentives for an increase in energy efficiency, so that less energy input is required to obtain the same level of energy services. More importantly, however, the carbon price causes a higher electrification of energy consumption and a switch to non-fossil-fuel power in the production of electricity.

Continued growth requires a scaling up and substantial reallocation of energy-related investments. Investment follows the changes in output closely and reflects the energy intensity of the sectors. Relative to the baseline, utilities increase investment strongly and absorb most of the reduction in investment in other sectors, especially in fossil-fuel extraction and EITE sectors. Overall, energy-related investment needs to increase by about 50 percent from current levels by 2030, relative to 25 percent in the baseline.

Changes in energy consumption

(Mtoe; percent change from 2019)



Changes in electricity mix

(TWh; deviation from 2019)

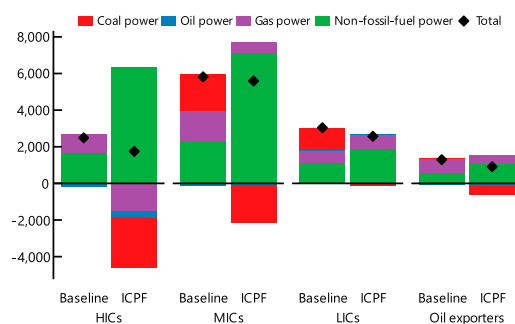


Figure 4. ICPF and baseline scenarios: Changes in energy consumption and electricity mix in 2030. Source: IMF-ENV model., Note: HIC = high-income country; ICPF = international carbon price floor; LIC = low-income country; MIC = middle-income country; Mtoe = Megatons of oil equivalent; TWh = Terawatt hours.

3.4 International burden sharing

To illustrate the efficiency and international burden-sharing properties of the ICPF, the study compares it with a global tax scenario. A uniform global carbon tax has long been suggested as the most efficient form of climate change mitigation at the global level. Thus, the study models it to compare it with the ICPF scenario, with its differentiated carbon prices. The global tax scenario is designed by calculating the total GHG emission reductions of the ICPF scenario and then setting a global uniform carbon price to match these emission reductions exactly. The endogenous carbon price in this scenario reaches US\$55.80/tCO₂eq in 2030 (2018 prices).

The ICPF has a less regressive distribution of emission reductions and GDP cost than the global tax scenario. [Figure 5](#) compares the emissions and GDP impacts of the ICPF and global tax scenarios across country groups (all measured in deviation from the baseline in 2030). Emission reductions are highest among HICs and lowest among LICs. In contrast, the global tax scenario implies much higher emission reductions among MICs and LICs than among HICs. The reason for this is that the emission intensity is highest at lower income levels; hence, for a given carbon price, countries reduce emissions by more. Broadly speaking, relative to the global tax scenario, the ICPF transfers a fraction of the emission reductions and the associated GDP cost from LICs to HICs, with MICs little affected since their carbon price floor under the ICPF is close to the uniform global carbon price.

The progressive distribution of emission reductions in the ICPF comes at the cost of less efficiency than the global tax scenario, but the loss in efficiency is limited to 0.3 percent of baseline GDP. The bottom panel of [Figure 5](#) illustrates the argument in favour of a uniform global carbon price: uniform pricing would imply lower GDP costs at the global level, at about 0.8 percent compared with 1.1 percent under the ICPF. The

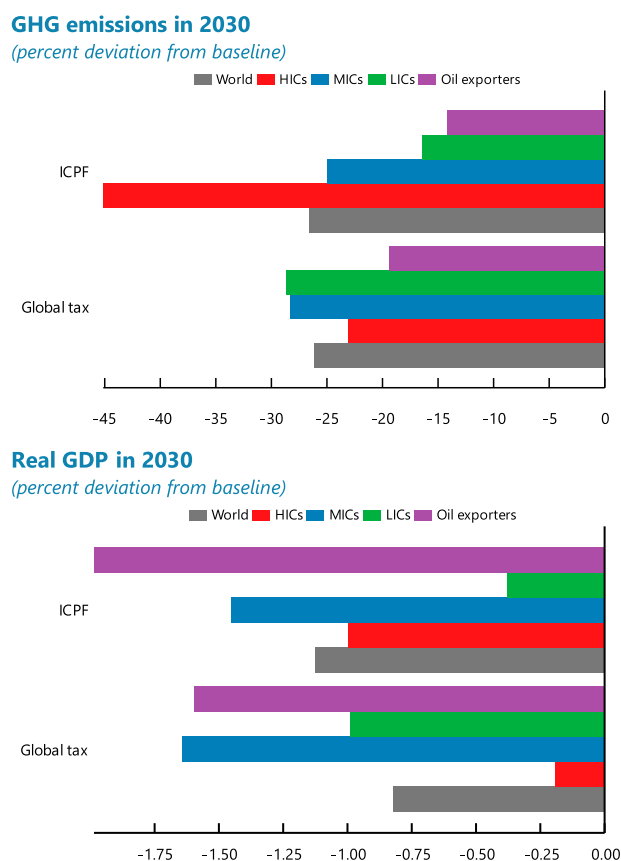


Figure 5. Emissions and GDP by aggregate country group.

reason is that it is most efficient to reduce emissions where abatement cost is lowest (in LICs, for example). The ICPF scenario implies greater GDP losses but a more progressive sharing of the mitigation costs burden than the global carbon tax scenario. This illustrates a traditional equity-efficiency trade-off.

The ICPF is progressive regarding emission reductions. It is thus an option for implementing the principle of common but differentiated responsibilities of the Paris Agreement. At the same time, relative GDP losses for MICs are higher than for HICs. The reason is that HICs can substitute away from emission-intensive activities more easily than MICs. Forcing a fully progressive GDP cost distribution through higher carbon prices for HICs would make the carbon price differential so large that HICs would be concerned about their competitiveness. Instead, the distribution of GDP cost could be made progressive through transfers. In fact, the Paris Agreement stipulates climate finance in Article 9. Adding the implementation of Article 9 to the ICPF would thus achieve progressivity in both variables. Given the focus of this paper on the effect of differentiated carbon pricing on the primary effect of burden sharing, we do not extend the modelling to carbon prices.

3.5 Competitiveness

Border carbon adjustment is intended to protect the implementing economy from a loss of competitiveness. The largest proposal to that effect is the carbon border adjustment mechanism (CBAM) of the European Union (European Commission, 2021). The protection of the domestic economy, however, is expected to come at a loss for trade partners. In the case of the EU, this will be China, Russia and India in particular (Zhong & Pei, 2022). As a result, it has been described as an unjustified trade barrier and as conflicting with rules of the World Trade Organization (Lim et al., 2021). To evaluate the competitiveness properties of the ICPF, we compare it to different versions of border carbon adjustment, as defined in Section 2.2. For a detailed description of BCA scenarios, see Appendix 3. Notice that among the scenarios compared in this sub-section, the ICPF has by far the largest emission reductions.

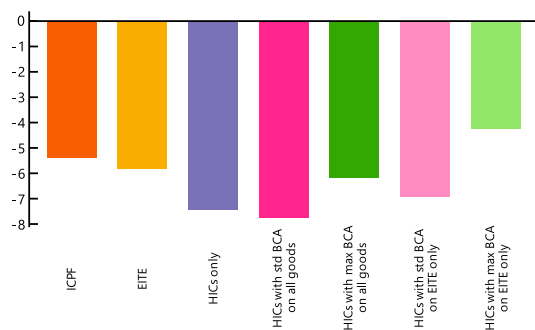
Our results confirm a number of conclusions from the literature on BCA, namely that (1) a border carbon adjustment helps reduce competitiveness losses of energy-intensive and trade-exposed sectors (Böhringer et al., 2022); (2) border carbon adjustments are not a perfect instrument for entirely restoring output levels of energy-intensive and trade-exposed industries (Thube et al., 2021); and (3) a border carbon adjustment shifts a small part of the burden of mitigation toward the noncoalition countries and sectors (Böhringer et al., 2012). We add two new findings: first, that an ICPF can be as effective as border carbon adjustment mechanisms at preserving competitiveness of high-income countries in energy-intensive and trade-exposed sectors. In addition, stronger forms of border carbon adjustments could offer incentives for the introduction of international carbon price floors by MICs and LICs in energy-intensive and trade-exposed sectors, in which concerns about losses of competitiveness for acting countries are the greatest.

The simulations show that an ICPF can be as effective as some designs of BCAs at protecting HICs against losses in competitiveness and market shares, despite allowing carbon prices to be differentiated by development levels. Figure 6 compares the impacts on gross output and global trade shares of EITE industries across the various scenarios. The ICPF performs better than the standard BCA and almost as well as the maximum BCA when it comes to limiting output losses of HICs in EITE sectors (like 'Iron and steel' and 'Non-ferrous metals') and changes in their world market shares, as shown in Figure 6. Indeed, while MICs and LICs benefit from lower carbon prices than HICs under the ICPF, the carbon intensity of their products is higher, evening out the carbon costs per unit of output.

This is also the case if a sectoral ICPF arrangement (EITE scenario) is considered. The outcomes for EITE sectors in HICs are very close to one another whether MICs and LICs impose the carbon price floor on their entire economies or on their EITE sectors only. It should thus be an acceptable protection of these sectors in HICs and make a BCA unnecessary. At the same time, MICs and LICs could be more amenable to such a sectoral ICPF. Indeed, Figure 7 shows that GDP costs of MICs and LICs would be broadly similar (or even slightly lower in some cases) with a sectoral ICPF than a BCA, because the tax would be lower (albeit it would apply to both exported and domestically sold goods) and they could keep the revenues from the carbon price. While this would not scale up mitigation efforts sufficiently, it would allow avoidance of the use of BCAs and provide a cooperative solution.

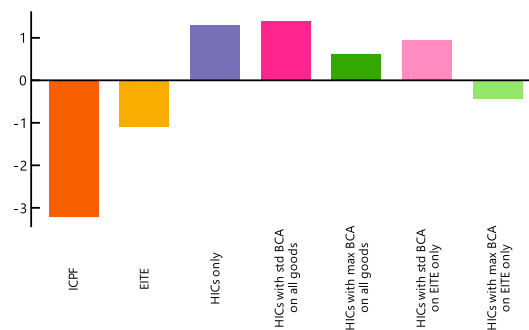
Real gross output of energy-intensive industries in high-income countries

(percent deviation from baseline)



Real gross output of energy-intensive industries in non-high-income countries

(percent deviation from baseline)



Global market share of energy-intensive industries¹

(percentage point deviation from baseline)

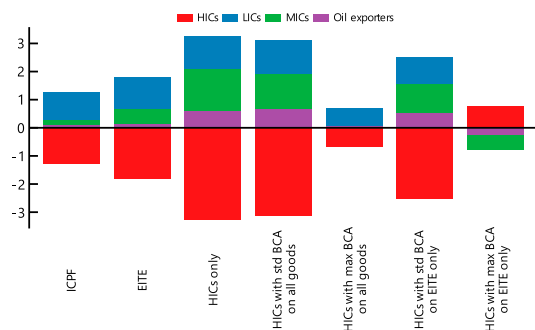
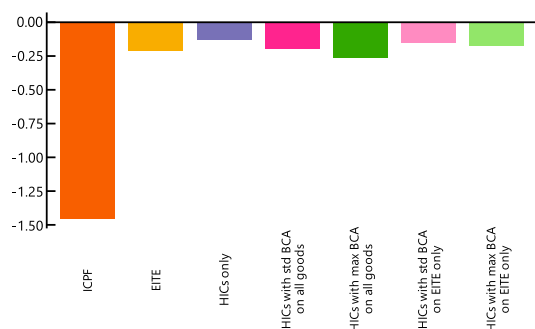


Figure 6. Partial-action scenarios: EITE industries' output and market shares in 2030. Source: IMF-ENV model. Note: BCA = border carbon adjustment; EITE = energy intensive and trade exposed; HIC = high-income country; ICPF = international carbon price floor; max = maximum; std = standard. ¹Market share for a given commodity is the value of exports of a country as a percentage of world total exports.

(percent change from baseline)

Middle-income countries



Low-income countries

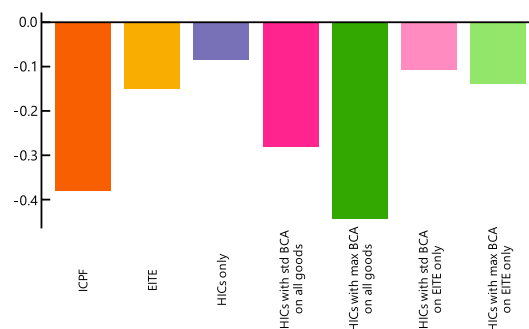


Figure 7. BCA and EITE scenarios: Real GDP in 2030 for middle- and low-income countries. Source: IMF-ENV model. Note: BCA = border carbon adjustment; EITE = energy intensive and trade exposed; HIC = high-income country; ICPF = international carbon price floor; max = maximum; std = standard.

4. Conclusions

An international carbon price floor arrangement would be an effective mechanism for scaling up global mitigation efforts towards what is required to keep within the temperature targets of the Paris Agreement. Current policies are not sufficient to stop global emissions from increasing, let alone to implement the substantial reductions that are required to reach net zero targets by midcentury. An international carbon price floor arrangement, as suggested by Parry et al. (2021), would help the global economy change course and head in the right direction. Key to such a mechanism is the concept of global minimum carbon prices – whether achieved explicitly by a carbon price or implicitly by an equivalent set of alternative policies – and the differentiation of these minimum carbon prices by level of development of countries.

We find that an international carbon price floor arrangement scales up climate mitigation at relatively small macroeconomic costs and is compatible with continued economic development. **Assuming that carbon-pricing revenues are used productively (in this paper, to reduce labour income taxes),** GDP growth remains very robust under the international carbon price floors and is only slightly lower than under the baseline. Growth projections for LICs in the decade 2021–2030, for example, would be reduced from a cumulative 58.3 percent in the baseline to 57.3 percent under the international carbon price floors. Underlying this continued growth is a substantial reallocation of energy-related investment from fossil fuels to non-fossil-fuel technologies. Investment in non-fossil-fuel power needs to increase by 168 percent from 2019 levels, and investment in fossil-fuel power to contract by 64 percent, by 2030.

A second result is that differentiated carbon prices under the international carbon price floor arrangement allow for more international burden sharing than a uniform carbon price, at a small efficiency cost. The international carbon price floor arrangement is progressive in terms of emissions reductions, as countries with higher income levels generally reduce their emissions proportionally more. This is also true for reductions in emissions per unit of GDP and emissions per capita. Compared with a uniform global carbon price, it shifts part of the emissions reductions and of the economic cost of mitigation from LICs to HICs. The economic efficiency, measured by the global GDP cost per unit of emission reduction, is a bit lower than under a uniform carbon price because the lowest-cost abatement options are in MICs and LICs, but only marginally so. However, despite much higher carbon prices and proportionally larger emissions reductions, HICs typically incur moderate mitigation costs, reflecting their very low emission intensity. This suggests a potential to complement carbon prices with cross-country transfers to increase further the international burden sharing. Oil exporting countries stand out, as their economic costs are above those of other countries, but these costs cannot be influenced much as they are driven by reduced demand for oil.

A third result is that the ICPF eliminates the need for border carbon adjustment. The main mechanism explaining this observation is the higher carbon content among producers from MICs and especially LICs. While they pay less per ton of CO₂, their products contain higher emission levels. As a result, the carbon tax paid per unit of output is similar across income groups.

While the first two results are in line with Bauer et al. (2020), this research validates these results in a different type of model (CGE), and applies the idea of differentiated policy stringency (and moderate transfers) to a concrete policy proposal of the IMF. In addition, the third result highlights that this idea also eliminates the need for border carbon adjustment, despite remaining differences in policy stringency. This analysis is based on the comparison of various policy designs to reduce losses in competitiveness. This line of research could be further advanced by incorporating recent empirical results, which reflect that carbon pricing causes no cost to GDP (Metcalfe & Stock, 2020) and that carbon leakage (Eskander & Fankhauser, 2023) as well as losses in competitiveness (Venmans et al., 2020) are estimated to be zero or close to zero.

Another contribution of this paper is the option to phase in an international carbon price floor arrangement through a sectoral version applied to the EITE sector only. This would address competitiveness concerns because losses in competitiveness are concentrated in EITE industries. At the same time, carbon pricing might be less controversial in EITE sectors. Protests against carbon prices have often been directed at carbon prices in transportation, as in the case of the yellow vests in France. Carbon pricing in the EITE sectors are less visible there than in other domestic energy consumption and transportation areas and thus may be more acceptable.

Note

1. The high prices are also a reflection of using carbon pricing as the only policy instrument in this analysis. Other policy instruments, such as regulations or targeted research and development to develop low-carbon technologies for hard-to-abate sectors could strongly reduce the need for high carbon prices. The international carbon price floor proposal allows for the use of alternative policies to carbon pricing and mixed approaches. Finally, anticipation effects – in which forward-looking agents front-load their adjustment to expected future carbon price increases – can also reduce the need for high up-front carbon prices (IMF, 2020).

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

IMF-ENV uses various data sources to calibrate both model equations and model projections.

The main source to calibrate the model is the GTAP-POWER database V10, including satellite accounts for CO₂ emissions, non-CO₂ emissions, air pollutants and energy. All this data is available at the GTAP center at <https://www.gtap.agecon.purdue.edu/databases/v10/index.aspx>.

The baseline calibration of the model makes use of macroeconomic projections built upon the IMF World Economic Outlook Database (October 2021) available at <https://www.imf.org/en/Publications/WEO/weo-database/2021/October> up to 2026. After this year macro data is extrapolated using (Dellink et al., 2017). These GDP, employment and current account projections are available upon request.

Population projections are from United Nation <https://population.un.org/wpp/>, using the 2019 Revision.

Energy projection used for calibrating long run energy system is a mix of the IEA World Energy Outlook projection for 2018 with some revisions from the 2020 projections, see <https://www.iea.org/reports/world-energy-outlook-2020> and <https://www.iea.org/reports/world-energy-outlook-2018>.

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